HENG BOON LONG

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE Aug 2022 - Present

Bachelor of Science in Business Analytics (School of Computing)

- Specialisation in Machine Learning-based Analytics
- GPA: 4.60 / 5.0 (On track for First-Class Honours)

INTERNSHIP & WORK EXPERIENCE

GOVTECH (SINGPASS) *Anti-fraud Data Scientist Intern* *Singapore* | *Aug 2025 – Present*

- **Experimented with deep learning models** on AWS SageMaker to improve Singpass's anti-fraud capability to detect suspicious login behaviour, escalating high-risks account to anti-fraud analysts and SPF.
- **Orchestrated MLOps pipelines** using AWS Glue with PySpark and Apache Airflow on Pulumi (IaaC) to process millions of Singpass login transactions daily, enabling batch ML predictions at scale.
- **Developed a Python script** to process over 7M+ ICA source photo, identifying low-quality photo with heuristic metrics and recommending targeted remediations to boost Singpass Face Verification accuracy.

POLICYME *Software Engineer / Data Scientist Intern* *Canada* | *Aug 2024 – Jul 2025*

(Worked as part of a year-long entrepreneurship programme at a technology-driven Insurtech scale-up in Canada)

- **Architected a distributed error monitoring system** across 20+ microservices using custom middleware and event filtering, reducing alert noise by 50% and improving incident response speed for engineering teams.
- **Created custom linting rules for React and Flask** to enforce consistent coding standards and seamless integration with performance monitoring platform (Sentry), raising code quality across teams.
- **Built analytics infrastructure** with DBT and BigQuery, enabling self-serve Looker dashboards for Engineering Managers to track release quality, sprint performance and bug resolution across Engineering teams.

AFFYN *Data Analytics Intern* *Singapore* | *Sep 2023 – Jan 2024*

- **Designed a scalable blockchain data ingestion pipeline** on Google Cloud services, processing 100K+ daily transactions into a centralised data warehouse for real-time analytics.
- **Developed a machine learning churn prediction model** using XGBoost and feature engineering techniques which led to a targeted retention campaign that improved user engagement by 15%.
- **Engaged in weekly discussions with product team** to review app performance metrics (stickiness, retention, DAU) and brainstorm solutions to improve user engagement.

TECHNICAL SKILLS

- **AI/ML & Deep Learning:** PyTorch | Scikit-Learn | NumPy | Pandas | XGBoost | NLTK
- **MLOps & Data Engineering:** Apache Airflow | PySpark | DBT | Docker | AWS Glue | AWS Sagemaker
- **Cloud Platforms & Infrastructure:** Google Cloud Platform | Amazon Web Services | Pulumi (IaaC)
- **Programming Languages:** Python | SQL | JavaScript | TypeScript | Java | R
- **Databases & BI / Visualization:** PostgreSQL | MySQL | AWS Redshift | BigQuery | Looker | Tableau

KEY PROJECTS

Empathetic Assistive RAG for Elderly Chatbot (Business Analytics Capstone Project) Aug 2025 – Nov 2025

(Created a RAG-driven elderly-assistive chatbot backend system for our client, Prescience Presbyrobotics, with natural language processing, memory routing, hybrid retrieval and a secure caregiver portal)

- **Designed and set up the PostgreSQL database on Neon** with pgvector extension, including ERD design, table schemas, security policies and pgcrypto encryption for healthcare and personal data.
- **Implemented full backend security architecture** for the caregiver interface using Flask, with JWT authentication, RBAC and audit logging to ensure compliant access to elderly records.
- **Developed all Flask backend endpoints** powering the caregiver portal (account provisioning, memory / healthcare updates, activity dashboards), integrating tightly with the encrypted cloud database.
- **Collaborated on the system-wide pipeline architecture in a team of 6 Engineers**, enabling clean module-to-module communication across Natural Language Normalization, Memory Routing, RAG retrieval and Caregiver Interface components.

Social Bot Detection on Twitter (Achieved Distinction in BT4222 Coursework) *Feb 2024 – May 2024*

- **Built Graph Neural Network models** combining user behavioural patterns and network analysis using PyTorch and NetworkX, achieving >90% accuracy across precision, recall and F1-score on the Twibot-22 benchmark.